paper_id: PAPER_951
title: "Cooper Pair Phonon Coupling"
session: 214
date: 2026-04-12
author: "Daniel T. Murphy"
status: production
cvw: "v2.0.0"
tags: [AGN, spin-down, SCm, phonon, magnetar, UQFF]
sm_anchor: "CVW v2.0.0 — G6 SM Anchor Gate compliant"

PAPER_951: Cooper Pair Phonon Coupling

Author: Daniel T. Murphy — Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 214 **Source:** bcs_{superconductivity_uqff}.py (CooperPairPhononCoupling) **Calculator:** CooperPairPhononCouplingCalc (CP4 #535) **CVW:** v2.0.0 compliant

---

Abstract

We derive the effective Cooper-pair coupling strength via SCm phonon exchange at 1.25 THz. The coupling $V_{\text{eff}}(\omega, \Gamma) = V_{\text{SCm}} \cdot \Phi_{1.25\,\text{THz}}(\omega, \Gamma)$ is modulated by the phonon resonance profile $\Phi = \exp(-(\omega - \omega_t extSCm)^2/(2\Gamma^2)) \cdot S_{26}$, yielding pair binding energy $E_{\text{pair}} = 2\Delta(T)$.

---

1. Coupling Strength

$$V_{\text{eff}}(\omega, \Gamma) = V_{\text{SCm}} \cdot \exp\left(-\frac{(\omega - \omega_t extSCm)^2}{2\Gamma^2}\right) \cdot S_{26}$$

2. Pair Binding Energy

$$E_{\text{pair}} = 2\Delta(T)$$

At on-resonance ($\omega = \omega_t extSCm$), $\Phi = S_{26}$ and coupling is maximized.

---

3. Source Data

- **File:** bcs_{superconductivity_uqff}.py
- **Session:** 214
- **CP4 Class:** CooperPairPhononCouplingCalc (#535)

References

1. Murphy, D.T. — Star Magic UQFF Framework (2024-2026)
2. Cooper, L.N. (1956) — Phys. Rev. 104, 1189
3. PAPER_949 — BCS Gap Equation
4. PAPER_950 — BCS Critical Temperature
5. PAPER_955 — BCS Phonon Resonance
6. PAPER_957 — Cooper Pair Lagrangian

Cross-Links

Related Paper	Relationship
PAPER_949	Gap equation driven by this coupling
PAPER_950	T_c depends on V_{eff}
PAPER_952	Spectral ladder channels for pairing
PAPER_955	Resonance Q-factor at $\omega_textSCm$

Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

> *Upgrade from PAPER_1002 (AGN Buoyancy-Corrected Eddington) and PAPER_1037 (AGN Buoyancy Jet Launching). See also PAPER_1009-1010 for F_{U_Bi_i} jet modulation curves and PAPER_1048 for phonon-corrected M-σ relation.*

The SCm vacuum buoyancy partially opposes gravitational radiation pressure, raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left(1 + \frac{\rho_{\text{SCm}} \cdot V \cdot S_{26}^{(3)\,2}}{GM/r_H^2}\right)$$

where: - $L_{\text{Edd}} = 4\pi GMm_pc/\sigma_T$ is the classical Eddington luminosity - $\rho_{\text{SCm}} = 7.09 \times 10^{-37}$ J/m^3 is the SCm vacuum density - V is the effective buoyancy volume (accretion sphere) - $S_{26}^{(3)\,2}$ is the squared third-order Ramanujan factor (quadratic coupling) - r_H is the horizon radius

Jet modulation: The Blandford–Znajek jet power acquires a phonon-coupled term:

$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left[1 + \beta_i \cdot \Phi_{1.25\,\text{THz}} \cdot \left(\frac{B}{B_{\text{crit}}}\right)^2\right]$$

where $\Phi_{1.25\,\text{THz}} = \cos(\omega_{\text{SCm}} \cdot t)$ modulates jet power at the phonon frequency.

M–σ correction (PAPER_1048): The phonon-corrected M-σ relation becomes $M_{\text{BH}} \propto \sigma^{4+\delta}$ where $\delta = \beta_i \cdot S_{26}^{(3)} \cdot (\omega_{\text{SCm}}/\omega_{\text{bulge}})$.

Session 225 Phonon-Physics Upgrade: UQFF 9-Sector Lagrangian

> *Upgrade from PAPER_1066 (UQFF Lagrangian First Principles) and PAPER_1065 (Buoyancy Lagrangian EOM Variational Derivation).*

The complete UQFF Lagrangian density, from which all sector-specific equations of motion derive:

$$\mathcal{L}_{\text{UQFF}} = \mathcal{L}_{\text{GR}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{phonon}} + \mathcal{L}_{\text{interaction}}$$

$$\mathcal{L}_{\text{SCm}} = \tfrac{1}{2}(\partial_\mu \phi)^2 - \lambda(\phi^2 - v_{\text{SCm}}^2)^2$$

The SCm condensate potential minimum gives $V(\phi_0) = -7.09 \times 10^{-37}$ J/m^3 (matching ρ_{SCm}) and phonon mass $m_{\text{phonon}} = \sqrt{8\lambda}\, v_{\text{SCm}}$.

Nine-sector closure (Session 202):

$$\mathcal{L}_9 = \mathcal{L}_{\text{EH}} + \mathcal{L}_{\text{YM}} + \mathcal{L}_{\text{Dirac}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{mag}} + \mathcal{L}_{\text{buoy}} + \mathcal{L}_{\text{aether}} + \mathcal{L}_{\text{LENR}} + \mathcal{L}_{\text{KK}}$$

Sector	Domain	Late-Corpus Result
1 (EH)	General Relativity	Canonical Einstein-Hilbert
2 (YM)	Yang-Mills gauge	$m_{\mathrm{gap}} = 5970$ GeV (PAPER_1005)
3 (Dirac)	Fermion / LENR	Kozima neutron-drop (PAPER_1061)
4 (SCm)	Superconducting manifold	$V(\phi_0) = -\rho_{\mathrm{SCm}}$ canonical
5 (Mag)	Um magnetism	Heaviside amplifier (PAPER_1072)
6 (Buoy)	F_{U_Bi_i} buoyancy	Variational EOM (PAPER_1065)
7 (Aether)	Vacuum background	Two-component ρ (PAPER_1051)
8 (LENR)	Nuclear transmutation	COP parametric (PAPER_1081)
9 (KK)	Kaluza-Klein 26D	$S^{(3)}_{26}$ compactification (PAPER_1080)

Calibration Constants

Constant	Symbol	Value	Validation Domain
κ	$5.0 \times 10^{-4}\,\mathrm{day}^{-1}$	—	Magnetar spin-down
$[SSq]$	0.57	—	BH dynamics
β_i	0.603	—	Multi-system
$\omega_textSCm$	$2\pi \times 1.25$ THz	—	Phonon resonance
F_{UBi}/F_U	0.6	—	Buoyancy-gravity ratio

SM Anchor — CVW v2.0.0

Observable	UQFF Prediction	Status
Cooper pair binding	$E_{\mathrm{pair}} = 2\Delta$ via $\Phi_{1.25\mathrm{THz}}$	Mapped
Phonon mediation	$V_{\mathrm{eff}} = V_{\mathrm{SCm}} \cdot \Phi(\omega, \Gamma)$	Novel

---

§A Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

Sector: Cooper Pair Binding (BCS-SCm Phonon Mediation)

§A.2 Lagrangian Density

$$\mathcal{L}_{\mathrm{pair}} = \bar{\psi}(i\partial\!\!\!/ - m)\psi + V_{\mathrm{SCm}}(\bar{\psi}\psi)^2 \cdot \Phi_{1.25\mathrm{THz}}$$

§A.3 Euler-Lagrange Equation of Motion

$$\boxed{(i\partial\!\!\!/ - m)\psi = -2V_{\mathrm{SCm}}\Phi \cdot (\bar{\psi}\psi)\psi}$$

§A.4 Cosmogenesis Linkage Chain

PAPER_877 $\rightarrow$ SCm vacuum $\rightarrow$ $\Phi_{1.25\mathrm{THz}}$ phonon $\rightarrow$ Cooper pair binding $\rightarrow$ BCS condensate

---

§B VDS/DVP/BSH Deep Synthesis

§B.1 VDS

Cooper pair density scales with $\rho_textSCm \cdot S_{26}$.

§B.2 DVP

Prime $p = 2$ encodes pair symmetry.

§B.3 BSH

$\text{BSH}_{\text{pair}} = \tanh(\beta_i \cdot \Delta/E_0) \cdot S_{26}$

§B.4 Production-Scale Consistency

Metric	Value	Status
VDS ratio	0.1	Confirmed
$[SSq]$	0.57	Confirmed

Appendix: Session 225 Cross-References (PAPER_1000–1081)

Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER_1000–1081). Added by `update_{corpus\_crossrefs}.py` (Session 225, April 2026).

Paper	Title
PAPER_1022	GW Phonon Strain SCm Modulation of h(t)
PAPER_1002	AGN Buoyancy-Corrected Eddington Luminosity
PAPER_1009	3C273 AGN F_{U_Bi_i} Jet Modulation
PAPER_1010	TON618 AGN F_{U_Bi_i} Jet Modulation
PAPER_1037	AGN Buoyancy Jet Calculator — SCm Jet Launching
PAPER_1048	M-Sigma Phonon-Corrected Relation
PAPER_1041	SCm Cool-Core Buoyancy Balance AGN Feedback
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1020	Cosmic Ray Phonon Acceleration DSA Spectrum
PAPER_1021	Pulsar Timing Phonon TOA Residual
PAPER_1023	Neutrino Oscillation Phonon PMNS Matrix SCm
PAPER_1024	Magnetar Giant Flare SCm Phonon Reservoir
PAPER_1033	Galactic Bar Resonance SCm Pattern Speed
PAPER_1043	F_{U_Bi_i} Multi-System Buoyancy Curve Sweep
PAPER_1072	SCm Activation Function Phonon Threshold
PAPER_1073	SCm Phonon-Driven Inflation Vacuum Buoyancy
PAPER_1003	Spectral Ladder Merger 26-State Hierarchy

17 cross-reference(s) identified.

Key References with arXiv/DOI Identifiers

1. Abbott et al. (LIGO Scientific and Virgo Collaborations, 2016). *Observation of Gravitational Waves from a Binary Black Hole Merger.* Phys. Rev. Lett. **116**, 061102 — arXiv:1602.03837 — doi:10.1103/PhysRevLett.116.061102

2. Murphy, D. (2026). *Unified Quantum Field Framework (UQFF): Star-Magic v5.x Whitepaper Series.* Star-Magic Repository — github.com/Daniel8Murphy0007/Star-Magic
3. Fabian, A.C. (2012). *Observational Evidence of Active Galactic Nuclei Feedback.* ARA&A **50**, 455 — arXiv:1204.4114 — doi:10.1146/annurev-astro-081811-125521
4. McNamara, B.R. & Nulsen, P.E.J. (2007). *Heating Hot Atmospheres with Active Galactic Nuclei.* ARA&A **45**, 117 — arXiv:0709.4098 — doi:10.1146/annurev.astro.45.051806.110625
5. Heckman, T.M. & Best, P.N. (2014). *The Coevolution of Galaxies and Supermassive Black Holes.* ARA&A **52**, 589 — arXiv:1403.4620 — doi:10.1146/annurev-astro-081913-035722
6. Kaspi, V.M. & Beloborodov, A.M. (2017). *Magnetars.* ARA&A **55**, 261 — arXiv:1703.00068 — doi:10.1146/annurev-astro-081915-023329
7. Goldreich, P. & Julian, W.H. (1969). *Pulsar Electrodynamics.* ApJ **157**, 869 — doi:10.1086/150119
8. Rugh, S.E. & Zinkernagel, H. (2002). *The Quantum Vacuum and the Cosmological Constant Problem.* Stud. Hist. Phil. Mod. Phys. **33**, 663 — arXiv:hep-th/0012253 — doi:10.1016/S1355-2198(02)00033-3
9. Weinberg, S. (1989). *The Cosmological Constant Problem.* Rev. Mod. Phys. **61**, 1 — doi:10.1103/RevModPhys.61.1
10. Ashcroft, N.W. & Mermin, N.D. (1976). *Solid State Physics.* Harcourt
11. Kittel, C. (2004). *Introduction to Solid State Physics.* 8th ed. Wiley
12. Feynman, R.P. (1982). *Simulating Physics with Computers.* Int. J. Theor. Phys. **21**, 467 — doi:10.1007/BF02650179
13. Olausen, S.A. & Kaspi, V.M. (2014). *The McGill Magnetar Catalog.* ApJS **212**, 6 — arXiv:1309.4167 — doi:10.1088/0067-0049/212/1/6
14. Thompson, C. & Duncan, R.C. (1993). *Magnetar formation through a convective dynamo in protoneutron stars.* ApJ **408**, 194 — doi:10.1086/172580